PANJAB UNIVERSITY, CHANDIGARH

Outlines of tests, syllabi and courses of reading for postgraduate diploma in theoretical foundations of cognitive sciences (self-financed), (semester system) examinations, ~~2019-20~~ 2020-2021

Shivani Singh

Semester 1

	Subject Code	Subject
1	PGDCSHF 101	CONSCIOUSNESS STUDIES
2	PGDCSHF 102	DATA SCIENCE AND PSYCHOPHYSICS I
3	PGDTCS 103	EXPERIMENTATION IN COGNITIVE SCIENCES
4	PGDCSHF 104	COGNITIVE ANALYTICS WITH STATISTICS
5	PGDCSHF 105	PRINCIPLES OF NEUROSCIENCE

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Semester 2

	Subject Code	Subject
1	PGDCSHF 106	HUMAN COGNITIVE PROCESSES
2	PGDCSHF 107	NEUROPHENOMNEOLOGY
3	PGDCSHF 108	DATA SCIENCE AND PSYCHOPHYSICS II
4	PGDCSHF 109	COGNITIVE ERGONOMICS AND HUMAN FACTORS
5	PGDCSHF 110	INDUSTRY PROJECT

NOTE: There will be five theory papers each carrying 100 marks (Int: 20, Ext:80) in the semester and four theory papers each carrying 100 marks (Int: 20, Ext:80) in the Second se

Description of Courses and Skills Learned

Semester I

SUBJECTS	DESCRIPTION	Skills
CONSCIOUSNESS STUDIES	This course will help you to understand the various philosophical abstractions, which can aid in understanding fields of Cognitive Sciences and Consciousness Studies from both Western and Indian Perspective	• Critical Thinking • Philosophical Analysis • Scientific Analysis
DATA SCIENCE AND PSYCHOPHYSICS I	In this course, you will learn how to build programs and use them for applications in Data Sciences and Cognitive Sciences.	• Psychophysics • Data Science • Python
EXPERIMENTATION IN COGNITIVE SCIENCES	This course will help you to understand the methodology and implementation of the Scientific method in Cognitive Sciences via Psychophysics experimentation	• Scientific Method Execution • Psychometric Assessment • Psychophysics Experimentation
COGNITIVE ANALYTICS WITH STATISTICS	In this course, you will learn about the foundations of research and develop skills to analyze the data and make inferences from the analysis to inform build models about cognition & behavior	• Cognition Modelling • Behavior Modelling
PRINCIPLES OF NEUROSCIENCE	This course will help you to understand how the Human Brain is structured and how it functions. The course will focus on how Human Attention and Perception are implemented in Brain structures	• Brain Structure and Functioning • Brain Cell / Neuron Recording

Description of Courses and Skills Learned

Semester II

SUBJECTS	DESCRIPTION	Skills
HUMAN COGNITIVE PROCESSES	This course will help you to understand various human cognitive processes and their limitations like Perception, Object Recognition, Attention, and Thinking	• Manipulation of Human Cognition
NEUROPHENOMNEOLOGY	In this course, you will learn about various cognitive processes from Phenomenological, Cognitive and Neural Levels. Also, you will learn the usage of Meditation techniques in Neurophenomenology.	• Neuro-clinical Application • Brain Imaging Technologies • Methodology
DATA SCIENCE AND PSYCHOPHYSICS II	In this course, you will learn how to build advanced programs & Database and use it for applications in Data Sciences and Cognitive Sciences.	• Classical and Adaptive Psychophysical methods • E-Prime • Data Science
COGNITIVE ERGONOMICS AND HUMAN FACTORS	In this course, you will learn about various cognitive models, with their application in interface/experience/ergonomics and human factor designing and optimization.	• Cognitive User Experience and Interface Designing • Ergonomics Modelling and Designing
Industry Project	-	• In-hand experience

POST GRADUATE DIPLOMA IN COGNITIVE SCIENCES AND HUMAN FACTORS

FIRST SEMESTER

PGDCSHF 101: CONSCIOUSNESS STUDIES Marks 100
(Int: 20, Ext: 80)

Objective:

Instructions for the Paper-setter and candidates:

(i) The theory question paper will be 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment. The paper setter must put note (ii) in the question paper.

(iii) There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus to be answerec in about 25 to 30 words. The candidate is required to attempt any 10 short answer type questions i.e. 2 marks of each. It shall carry 20 marks and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all

UNIT – I

Western Philosophy Approach: Introduction to Philosophical analysis, Scientific Method, Overview of Philosophy of Mind, Behaviourism, Computationalism, Functionalism, Embodiment, Phenomenology, Cognitive Revolution, Interdisciplinary perspective

UNIT – II

Western Philosophy Approach: Philosophical assumptions of Cognitive Sciences, Mind-body problem, Free-will and Determinism, Mystery of Consciousness

UNIT – III

Indian Philosophical Approach:

Mind in Yoga and Samkhya, Consciousness in Yoga and Samkhya

Mind in Vedanta, Consciousness in Vedanta

UNIT – IV

Indian Philosophical Approach:

Mind in Buddhism, Consciousness in Buddhism

Mind in Kashmir Shaivism, Consciousness in Kashmir Shaivism

ESSENTIAL READINGS:

1. Thagard, Paul, Mind: Introduction to Cognitive Science, Published February 4th, 2005 by Bradford Bock (first published June 1st, 1996)
2. Thagard, Paul, Conceptual Revolution, Princeton University Press; December 14, 1992
3. Thagard, Paul, Computational Philosophy of Science, MIT Press, 1993
4. S. N. Dasgupta, History of Indian Philosophy, (series), Orient Book Distributors, 1975

ADDITIONAL READING:

5. Carruthers, Peter. (2004). Practical reasoning in a modular mind. Mind & Language,19, 259–278.
6. Barrett, H. Clark: Evolutionary psychology. In Frankish & Ramsey, Ch 13.
7. Spelke, Elizabeth. (1998). Nativism, empiricism, and the origins of knowledge.Infant Behavior and Development, 21, 181–200.
8. Shallice, T. (1988). From Neuropsychology to Mental Structure. Cambridge: Cambridge University Press. Chapters 10 & 11.
9. Davies, M. (2010). Double dissociation: Understanding its role in cognitive neuropsychology. Mind & Language, 25, 500-540.
10. Plaut, D. C. (1995). Double dissociation without modularity: evidence from connectionist neuropsychology. Journal of Clinical and Experimental Neuropsychology, 17, 291-321.
11. Van Orden, G. C., Pennington, B. F., and Stone, G. O. (2001). What do double dissociations prove? Inductive methods and isolable systems. Cognitive Science, 25, 111172.
12. Wetzels, R., Matzke, D., Lee, M. D., Rouder, J. N., Iverson, G. J., & Wagenmakers, E. J. (2011). Statistical evidence in experimental psychology an empirical comparison using 855 t-tests. Perspectives on Psychological Science, 6, 291-298.
13. Pashler, H., & Harris, C. R. (2012). Is the replicability crisis overblown? Three arguments examined. Perspectives on Psychological Science, 7, 531-536.
14. Meehl, P. E. (1990). Why the summaries of research on psychological theories are often uninterpretable. Psychological Reports, 66, 195-244.
15. Stegenga, J. (2011). Is meta-analysis the platinum standard of evidence? Studies in history and philosophy of science part C: Studies in history and philosophy of biological and biomedical sciences, 42, 497-507

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16. Klein, C. (2010). Images are not the evidence in neuroimaging. The British Journal for the Philosophy of Science, 61, 265-278.
17. Frankenhuis, Willem & Ploeger, Annemie. (2007). Evolutionary psychology versus Fodor: Arguments for and against the massive modularity hypothesis. Philosophical Psychology, 20, 687–710.
18. Glymour, C. (1994). On the methods of cognitive neuropsychology. The British journal for the philosophy of science, 45, 815-835.
19. Machery, E. (2012). Dissociations in neuropsychology and cognitive neuroscience. Philosophy of Science, 79, 490-518.
20. Poldrack, R. A. (2006). Can cognitive processes be inferred from neuroimaging data? Trends in Cognitive Sciences, 10, 59–63.
21. Prinz, Jesse. (2004). Furnishing the mind, MIT Press.
22. Samuels, Richard. (2004). Innateness in cognitive science. Trends in Cognitive Sciences, 8, 136–141
23. Machery, E. (2012). Dissociations in neuropsychology and cognitive neuroscience. Philosophy of Science, 79, 490-518.

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POST GRADUATE DIPLOMA IN COGNITIVE SCIENCES AND HUMAN FACTORS

FIRST SEMESTER

PGDTFCS 102: DATA SCIENCE AND PSYCHOPHYSICS I

Marks 100

(Int: 20, Ext: 80)

Objective:

Instructions for the Paper-setter and candidates:

(i) The theory question paper will be 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment. The paper setter must put note (ii) in the question paper.

(iii) There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words. The candidate is required to attempt any 10 short answer type questions i.e. 2 marks of each. It shall carry 20 marks and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all

UNIT – I

Defining Data Science, R vs Python, Data Science Tools and Technologies, Regression, Application of Data Science, Data Science in Business, Use Cases of Data Science, The Report Structure, Data Scientist

UNIT – II

The Context of Software Development, Development Tools, Learning Programming with Python, Interactive Shell, IBM Watson Studio

Types, Expression and Variables, String operations, Lists, Tuples, Sets, Dictionaries

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UNIT – III

Conditioning and Branching, Loops, Functions, Objects, and Classes

Reading and Writing Files, loading data with Pandas, Working and Saving data in Pandas

Minor Project

UNIT – IV

Classical Psychophysics: Tasks, Forced Choice, Non-Forced Choice, Criterion-free, Criterion-dependent, Detection, Discrimination, Threshold, Eprime framework, and implementation

ESSENTIAL READINGS:

1. Kingdom, F. A., & Prins, N. (2016). Psychophysics: A Practical Introduction. Cambridge, MA: Academic Press.
2. Liu, Y. (2017). Python Machine Learning By Example. Birmingham, England: Packt Publishing.
3. Thanaki, J. (2017). Python Natural Language Processing. Birmingham, England: Packt Publishing.

ADDITIONAL READING:

1. Cannon, J. (2014). Python Programming for Beginners: An Introduction to the Python Computer Language and Computer Programming. Scotts Valley, CA: CreateSpace.
2. Chopra, D., Joshi, N., & Mathur, I. (2016). Mastering Natural Language Processing with Python. Birmingham, England: Packt Publishing.
3. Kamthane, A.N., & Kamthane, A. A. (n.d.). Programming and Problem Solving with Python. New York, NY: McGraw-Hill Education.
4. Karumanchi, N. (2015). Data Structure and Algorithmic Thinking with Python. Careermonk Publications.
5. Lutz, M. (2013). Learning Python: Powerful Object-Oriented Programming. Sebastopol, CA: O'Reilly Media.

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POST GRADUATE DIPLOMA IN COGNITIVE SCIENCES AND HUMAN FACTORS

FIRST SEMESTER

PGDCSHF 103: EXPERIMENTATION IN COGNITIVE SCIENCES

Marks 100

(Int: 20, Ext: 80)

Objective:

Instructions for the Paper-setter and candidates:

(i) The theory question paper will be 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment. The paper setter must put note (ii) in the question paper.

(iii) There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words. The candidate is required to attempt any 10 short answer type questions i.e. 2 marks of each. It shall carry 20 marks and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all

UNIT – I

The Complexity of Cognition, The First Cognitive Scientists, Behavioural and Physiological Approaches to Cognition, Relationship between Brain activity and cognition, Psychometric testing (IQ and BIG Five)

CogLab: Psychometric testing (IQ and BIG Five), Reaction Time (RT), Mental Rotation

UNIT – II

Neurons, Neuronal Communication, Neuronal information Processing, Neural Code of Perceiving Faces

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UNIT – III

Bottom-Up and Top-Down Processing, Recognizing Letters and Objects, Gestalt Laws. Heuristics for Perceiving, Perception, and Attention

CogLab: Perceiving an Image, Word Superiority effect, Faces in Landscape, Shape from Shading, Strings of Sounds, Change Detection

UNIT – IV

Attention and Selection, Early Selection Model, Intermediate Selection Model, Late Selection Model, Automatic Processing, Divided Attention, Attention, and Eye Movement; Object-based Visual Attention

CogLab: Dichotic Listening, Flanker-Compatibility Task, Stroop effect, Eye movement detection, Pre-cueing

ESSENTIAL READINGS:

1. Dawson, M. R. (2013). *Mind, Body, World: Foundations of Cognitive Science.*
2. Eckardt, B. V. (1995). *What is Cognitive Science?* Cambridge, MA: MIT Press.
3. Jackendoff, R., & Jackendoff, R. S. (1990). *Consciousness and the Computational Mind.* Bradford Books.
4. Morris, R. G., Tarassenko, L., & Kenward, M. (2005). *Cognitive Systems - Information Processing Meets Brain Science.* Amsterdam, Netherlands: Elsevier.

ADDITIONAL READING:

1. Bechtel, W. (2012). Mental Mechanisms: Philosophical Perspectives on Cognitive Neuroscience. London, England: Psychology Press.
2. Ross, B. H., & Irwin, D. E. (2003). Cognitive Vision: Psychology of Learning and Motivation. Gulf Professional Publishing.

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FIRST SEMESTER

PGDCSHF 104: COGNITIVE ANALYTICS WITH STATISTICS

Marks 100

(Int: 20, Ext: 80)

Objective:

Instructions for the Paper-setter and candidates:

(i) The theory question paper will be 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment. The paper setter must put note (ii) in the question paper.

(iii) There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words. The candidate is required to attempt any 10 short answer type questions i.e. 2 marks of each. It shall carry 20 marks and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all

UNIT – I

Quantitative and Qualitative Research, Scientific Method, Hypothesis Testing, Population, Sample, Variables, Research Designs, Randomized Controlled Trails, Meta-analysis

SatsLab: Working with SPSS, Data Entry, Inputting Designs

UNIT – II

A measure of Central Tendency, Sampling Error, Graphically Describing Data, The Normal Distribution, Presenting Descriptive Statistics

SatsLab: Central Tendency, Graphical Representation Generation, Generating Scattergrams, Obtaining Measure of Variation, Displaying Normal Curve on Histogram

UNIT – III

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Applying Probabilities to Research: Hypothesis testing, Null hypothesis, The Significance Level, Statistical Significance, Interpretation of P-value, Statistical Tests, Type I & II Errors, Alpha – 0.05, One-Tailed and Two-Tailed Hypothesis, Underlying Assumptions of Statistical Tests

SatsLab: Obtaining Confidence Intervals, Error Bar Charts, Generating Error Bar Charts, Results Coach, Statistics Coach

UNIT – IV

Analysis of Two Conditions via t-test

Criterion Significance Levels, Effect Size, Power, Factors Influencing Power, Calculating Power and Confidence Intervals

SatsLab: Pearson's r, Scattergram matrix, Partial Correlations r, Independent t-test, Two Samples Repeated-Measure design Paired t-test, Calculating Power

ESSENTIAL READINGS:

1. Field, A. (2013). *Discovering Statistics Using IBM SPSS Statistics*. Thousand Oaks, CA: SAGE.
2. Gauvreau, K. (2000). *Principles of Biostatistics*. Pacific Grove, CA: Brooks/Cole.
3. Upton, G., & Cook, I. (1996). *Understanding Statistics*. New York, NY: Oxford University Press.

ADDITIONAL READING:

1. Motulsky, H. (2010). *Intuitive Biostatistics: A Nonmathematical Guide to Statistical Thinking*. New York, NY: Oxford University Press, USA.
2. Bartholomew, D. J. (2015). *Statistics without Mathematics*. Thousand Oaks, CA: SAGE

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POST GRADUATE DIPLOMA IN COGNITIVE SCIENCES AND HUMAN FACTORS

FIRST SEMESTER

PGDCSHF 105: PRINCIPLES OF NEUROSCIENCE

Marks 100

(Int: 20, Ext: 80)

Objective:

Instructions for the Paper-setter and candidates:

(i) The theory question paper will be 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment. The paper setter must put note (ii) in the question paper.

(iii) There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words. The candidate is required to attempt any 10 short answer type questions i.e. 2 marks of each. It shall carry 20 marks and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all

UNIT – I

The Brain and Behaviour, Nerve Cells and Behaviour, Genes and Behaviour, Cytology of Neurons, Synthesis and Trafficking of Neuronal Proteins, Ion Channels, Membrane Potentials, Local Signalling: Passive Electric Properties of the Neurons, Propagated Signalling: Action Potential

UNIT – II

Synaptic Transmission, Directly Gated Transmission, Synaptic Integration, Modulation, Transmitter Release, Neurotransmitters

UNIT – III

The Anatomical Organization of the Central Nervous System, Functional Organization of Perception and Movement, Cerebral Cortex and Cognition, Cellular Representation for Perception and Action

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UNIT – IV

Coding of Sensory Information, Touch, Pain, Visual Image, Visual Processing by Retina, Central Visual Pathways, Perception of Motion, Depth and Forms, Colour Vision, Hearing, Smell and Taste, Motivational and Addictive States

NeuroLab: Single Cell Recording, Muscle Firing

ESSENTIAL READINGS:

1. Feng, J. (2003). *Computational Neuroscience: A Comprehensive Approach*. Boca Raton, FL: CRC Press.
2. Purves, D. (2012). *Neuroscience*. Sunderland, MA Sinauer Associates.
3. Revest, P., & Longstaff, A. (1998). *Molecular Neuroscience*. London, England: Garland Science.
4. Scott, A. (2007). *Neuroscience: A Mathematical Primer*. Berlin, Germany: Springer Science & Business Media.

ADDITIONAL READING:

1. Buccafusco, J. J. (2000). *Methods of Behavior Analysis in Neuroscience*. Boca Raton, FL: CRC Press.
2. Gross, C. G. (1999). *Brain, Vision, Memory: Tales in the History of Neuroscience*. Cambridge, MA: MIT Press.

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POST GRADUATE DIPLOMA IN COGNITIVE SCIENCES AND HUMAN FACTORS

SECOND SEMESTER

PGDCSHF 106: HUMAN COGNITIVE PROCESSES **Marks 100**

(Int: 20, Ext: 80)

Objective:

Instructions for the Paper-setter and candidates:

(i) The theory question paper will be 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment. The paper setter must put note (ii) in the question paper.

(iii) There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words. The candidate is required to attempt any 10 short answer type questions i.e. 2 marks of each. It shall carry 20 marks and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all

UNIT – I

Perception: Perceptual Constancy, Depth Perception, Visual Illusions, Theoretical Explanations of Illusions, Marrs theory, Primal Sketch, 2.5-D Representation, 3-D Model, Evaluation of Marrs Approach

UNIT – II

Object Recognition: Templates and Prototype, Feature Analysis, Pandemonium, Context Effect, Local & Global Relationship

UNIT – III

Attention: The Cocktail Phenomenon, Visual Studies, Inhibitory Mechanisms, Attentional Unification, Spotlight Model of Visual Attention: Visual Orienting, Neurological impairments of Orienting, Central and Peripheral Cueing, Zoom-Lens Modification, Inhibition of return

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UNIT – IV

Thinking: Problem-Space Theory, Deductive Reasoning, Mental Models, Inductive Reasoning hypothetico-deductive reasoning, Pragmatic Reasoning Schemata, Statistical reasoning

Disorders in Thinking: Case Studies of AP, DN, and FS

ESSENTIAL READINGS:

1. International Symposium on Attention and Performance. (2000). *Control of Cognitive Processes: Attention and Performance XVIII*. Cambridge, MA: MIT Press.
2. Malim, T. (1994). *Cognitive Processes: Attention, Perception, Memory, Thinking and Language*. London, United Kingdom: Macmillan International Higher Education.
3. Ohta, N., MacLeod, C. M., & Uttl, B. (2006). *Dynamic Cognitive Processes*. Berlin, Germany: Springer Science & Business Media.
4. Serra, R., & Zanarini, G. (2013). *Complex Systems and Cognitive Processes*. Berlin, Germany: Springer Science & Business Media.

ADDITIONAL READING:

1. Domaneschi, F. (2016). *Presuppositions and Cognitive Processes: Understanding the Information Taken for Granted*. Basingstoke, England: Springer.
2. Groome, D., & Groome, D. O. (2013). *An Introduction to Cognitive Psychology: Processes and Disorders*. London, England: Psychology Press.

UNIT – III

Brain Imaging Techniques: fMRI, EEG

Neurophenomenology through Meditation and Mindfulness

UNIT – IV

Neurophenomenological Reciprocal constraints, Neurodynamical approach

ESSENTIAL READINGS:

1. Gallagher, S., Janz, B., Reinerman, L., Trempler, J., & Bockelman, P. (2015). *A Neurophenomenology of Awe and Wonder: Towards a Non-Reductionist Cognitive Science*. Basingstoke, England: Springer.
2. Gazzaniga, M., & Ivry, R. (2013). *Cognitive Neuroscience: The Biology of the Mind: Fourth International Student Edition*. W.W.Norton.
3. Gordon, S. (2013). *Neurophenomenology and Its Applications to Psychology*. Berlin, Germany: Springer Science & Business Media.
4. Johns, P. (2014). *Clinical Neuroscience: An Illustrated Colour Text*. London, England: Churchill Livingstone.

ADDITIONAL READING:

1. Graham, G. (2013). *The Disordered Mind: An Introduction to Philosophy of Mind and Mental Illness*. London, England: Routledge.
2. Young, P. A., Young, P. H., & Tolbert, D. L. (2008). *Basic Clinical Neuroscience*. Philadelphia, PA: Lippincott Williams & Wilkins.

ESSENTIAL READINGS:

1. Kingdom, F. A., & Prins, N. (2016). Psychophysics: A Practical Introduction. Cambridge, MA: Academic Press.
2. Liu, Y. (2017). Python Machine Learning By Example. Birmingham, England: Packt Publishing.
3. Thanaki, J. (2017). Python Natural Language Processing. Birmingham, England: Packt Publishing.

ADDITIONAL READING:

1. Cannon, J. (2014). Python Programming for Beginners: An Introduction to the Python Computer Language and Computer Programming. Scotts Valley, CA: CreateSpace.
2. Chopra, D., Joshi, N., & Mathur, I. (2016). Mastering Natural Language Processing with Python. Birmingham, England: Packt Publishing.
3. Kamthane, A. N., & Kamthane, A. A. (n.d.). Programming and Problem Solving with Python. New York, NY: McGraw-Hill Education.
4. Karumanchi, N. (2015). Data Structure and Algorithmic Thinking with Python. CareerMonk Publications.
5. Lutz, M. (2013). Learning Python: Powerful Object-Oriented Programming. Sebastopol, CA: O'Reilly Media.

POST GRADUATE DIPLOMA IN COGNITIVE SCIENCES AND HUMAN FACTORS

SECOND SEMESTER

PGDCSHF 108: DATA SCIENCE AND PSYCHOPHYSICS II

Marks 100

(Int: 20, Ext: 80)

Objective:

Instructions for the Paper-setter and candidates:

(i) The theory question paper will be 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment. The paper setter must put note (ii) in the question paper.

(iii) There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words. The candidate is required to attempt any 10 short answer type questions i.e. 2 marks of each. It shall carry 20 marks and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all

UNIT – I

Importing Database, Domain, Dataset, Python package for Data Science

Identifying and Handle Missing values, Data Formatting, Data Normalization, Binning, Indicator Variables

UNIT – II

Summarizing Data Frame, Modelling, Prediction and Decision making

UNIT – III

Model evaluation and Refinement

UNIT – IV

Experimental Design in E-Prime, Lists, Image, Feedback, SoundOut Objects, Movie, Wait Objects, Labels, Experiment properties, Nested List

CogLab: The Simon Task, Implicit Association Task

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POST GRADUATE DIPLOMA IN COGNITIVE SCIENCES AND HUMAN FACTORS

SECOND SEMESTER

PGDCSHF 109: COGNITIVE ERGONOMICS AND HUMAN FACTORS

Marks 100

(Int: 20, Ext: 80)

Objective:

Instructions for the Paper-setter and candidates:

(i) The theory question paper will be 80 marks and 20 marks will be for internal assessment.

(ii) For private candidates, who have not been assessed earlier for internal assessment, the marks secured by them in theory paper will proportionately be increased to maximum marks of the paper in lieu of internal assessment. The paper setter must put note (ii) in the question paper.

(iii) There shall be 9 questions in all. The first question shall be short answer type containing 15 short questions spread over the whole syllabus to be answered in about 25 to 30 words. The candidate is required to attempt any 10 short answer type questions i.e. 2 marks each. It shall carry 20 marks and shall be compulsory question. Rest of the paper shall contain 4 units. Each unit shall have two questions and the candidates shall be given internal choice i.e. the candidates shall attempt one question from each Unit – 4 in all

UNIT – I

Ergonomics in work systems, Modern ergonomics, Methods: Ethnographic analysis, Focus Groups, Iterative design, Meta-analysis, Survey, Task Analysis, Motion Capture

UNIT – II

History of Ergonomics, History of Human-Computer Interaction, Human-Computer Interaction (HCI): Human-centred design process for interactive systems, Designing information in external memory stores, Human-computer dialogues

UNIT – III

Memory and Language in everyday life, effectiveness, and cost-effectiveness, Research issue, Human error and equipment designs, Cognitive Load in human-machine interaction

UNIT – IV

Cognitive aspects of human error, Characterising human-machine interactions, GOMS, Prevention of error in human-machine interaction, Psychosocial factors, Cross-cultural considerations

ESSENTIAL READINGS:

1. Bridger, R. (2008). *Introduction to Ergonomics, Second Edition*. Boca Raton, FL: CRC Press.
2. Forsythe, C., Liao, H., Trumbo, M. C., & Cardona-Rivera, R. E. (2014). Cognitive Neuroscience of Human Systems: Work and Everyday *Life*. Boca Raton, FL: CRC Press.

ADDITIONAL READING:

1. Hollnagel, E. (2003). *Handbook of Cognitive Task Design*. Boca Raton, FL: CRC Press.
2. Vu, K. L., & Proctor, R. W. (2011). *Handbook of Human Factors in Web Design, Second Edition*. Boca Raton, FL: CRC Press.